Positivity of Cylindric Kostka Polynomials via Lattice Path Crossings and the Franklin Involution

Robin Langer

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Abstract

We prove that the cylindric Kostka polynomials $Q_{n,c}(q)$ arising from the rank-3 generating function of cylindric plane partitions have non-negative integer coefficients, confirming a conjecture of Warnaar (Conjecture 2.7) and Corteel–Dousse–Uncu (Conjecture 4.2) in the case where the corrected coefficients h_m are non-negative ($d \not\equiv 0 \pmod 3$).

The proof proceeds in four stages. First, we reformulate the Q-polynomials in terms of Q-commutation relations between row and column operators on the ring of symmetric functions, and identify the local rules that realise these relations bijectively. Second, we interpret the ground set $\mathcal{G}_{n,c}$ as a space of lattice path families on the cylinder, where the crossing set S records positions at which paths intersect and the Euler factor $(zq;q)_n$ generates all possible crossing configurations. Third, we construct a sign-reversing, weight-preserving involution on $\mathcal{G}_{n,c}$ using a Franklin-style comparison between the peel level (the first position where a crossing can be introduced) and the glue level $\min(S)$ (the first existing crossing). The involution decomposes into n independent levels: at level k, element k is toggled in or out of S, pairing intersecting families with less-intersecting ones. After all n levels, only non-intersecting families ($S = \emptyset$) survive, each contributing with sign $+1$, giving $Q_{n,c}(q) \in \mathbb{N}[q]$.

Contents

I The Conjecture **2**

1 Background **3**

2 The Positivity Conjecture **4**

II The Algebraic Framework **5**

3 Q-Commutation Relations **6**

4 Q-Local Rules **7**

5 The Q-Schützenberger Identity **8**

III Lattice Paths and Crossings **9**

6 Lattice Paths on the Cylinder **10**

7 Q-Lattice Path Families and Crossings **11**

IV The Involution **12**

8 The Canonical Peel **13**

9 The Recursive Franklin Involution **14**

A Computational Evidence **15**

B The Cyclic Descent Injection **16**

C The $d \equiv 0 \pmod 3$ Obstruction **17**

Part I

The Conjecture

Chapter 1

Background

The story begins with three levels of structure that have appeared, in different guises, across three decades of work on cylindric partitions: the algebraic level (commutation relations between operators on symmetric functions), the combinatorial level (local rules applied within growth diagrams), and the analytic level (identities of Rogers–Ramanujan type).

The central theme is that each algebraic commutation relation between row and column operators has a precise bijective counterpart in the form of a local rule. This operator–bijection correspondence, developed across a Masters thesis [21], the cylindric partitions paper [22], and a PhD thesis [23], provides the toolkit for everything that follows.

Chapter 2

The Positivity Conjecture

We now state the objects and the conjecture precisely. The generating function $\mathrm{GK}_c(z,q)$ of rank-3 cylindric plane partitions with profile $c=(c_0,c_1,c_2)$ factors as

$$\mathrm{GK}_c(z,q)=\frac{1}{(zq;q)_\infty}\cdot F_c(z,q),$$

where $F_c=(zq;q)_\infty\cdot\mathrm{GK}_c$ is the "purely cylindric" part. The polynomials $Q_{n,c}(q)$ arise as

$$Q_{n,c}(q)=(q;q)_n\cdot[z^n]\,F_c(z,q).$$

Conjecture 2.1 (Warnaar [19]; Corteel–Dousse–Uncu [5])**.** *For all $n\geq 0$ and all rank-3 profiles c, $Q_{n,c}(q)\in\mathbb{N}[q]$.*

This chapter gives precise definitions of the four key polynomials — g_m, h_m, Q_n, and the Q-kernel $\Phi_{a,b}$ — records what is proved, what is computationally verified, and what remains conjectural, and illustrates with examples.

Part II

The Algebraic Framework

Chapter 3

Q-Commutation Relations

The commutation relations between row and column operators on Λ lift to the Q-enriched setting. At Level 3 in the hierarchy of generating functions, we obtain a genuine operator $\hat{Q}_{c,\mathbf{n}}$ on Λ whose trace recovers the multivariate cylindric Kostka polynomial. The adjoint $\hat{Q}^*$ corresponds to the reversed-complementary profile, and satisfies $\mathrm{tr}(\hat{Q}^*) = \mathrm{tr}(\hat{Q})$. A second operator $Q(z) = \sum Q_m z^m P_m$ packages the Q-polynomials as Pieri multiplication operators, revealing the cascade commutator structure that governs how Q-weights interact.

Chapter 4

Q-Local Rules

Each commutation relation has a bijective counterpart: a local rule applied within a growth diagram. The Burge rule (cascade up) and RSK rule (cascade down) are the two fundamental local rules; the Schützenberger involution relates them. In the Q-enriched setting, the Schützenberger involution *fails*: the Q-kernel $\Phi_{a,b}(t)$ is context-sensitive, so changing the cascade direction at one face propagates weight changes through the entire growth diagram. This failure is structurally informative — it shows that effective involutions for Q-positivity must be *local*, matching the recursive structure of the shifted factorisation $\Phi_{a,b}(t) = Q_a Q_b + t\,\Phi_{a+1,b+1}(t)$.

The Bender–Knuth involutions σ_i^Q provide the atomic local operations: each switches the cascade direction at the i-th diagonal of an ALCD. The condition $(\sigma_i^Q)^2 = \mathrm{id}$ holds by a direct case analysis on the local rule configurations.

Chapter 5

The Q-Schützenberger Identity

The Q-Schützenberger involution ψ_Q acts on the pair set $\mathcal{P}_{a,b,k} = \mathcal{G}_{a+k,c} \times \mathcal{G}_{b+k,c}$ as the composition of Q-Burge^{-1} (Burge-glue on the first component) with Q-RSK (RSK-peel on the second). This is *sign-preserving* and weight-preserving — it complements the sign-*reversing* involution ι. Together, ι and ψ_Q generate a dihedral group action on the pair set, giving the Q-Schützenberger identity: the fixed-point structure of the combined action constrains the positivity of the Q-kernel $\Phi_{a,b}(t)$.

The identity reveals the *orthogonal structure* of the positivity problem: ι handles sign cancellation (within a single ground set), while ψ_Q handles the coupling between two ground sets (the pair framework). Positivity of Q_n requires only ι; positivity of $\Phi_{a,b}$ requires both.

Part III

Lattice Paths and Crossings

Chapter 6

Lattice Paths on the Cylinder

Cylindric plane partitions biject with families of non-intersecting lattice paths (NILPs) on the cylinder. This chapter gives the classical model: lattice paths with North and East steps on a periodic strip, the cylindric closure condition, and the bijection CPP $\leftrightarrow$ NILP that translates the interlacing conditions of a CPP into the non-intersection constraint for lattice paths. The Q-kernel $\Phi_{a,b}(t)$ acquires a two-path interaction interpretation: it measures the interaction between paths a and b when they are forced to share the same strip.

Chapter 7

Q-Lattice Path Families and Crossings

This is the bridge between the algebraic framework (Parts I–II) and the involution (Part IV). A *Q-lattice path family* is a pair $(S, \mathfrak{p})$ where $\mathfrak{p}$ is a non-intersecting family and $S \subseteq \{1, \ldots, n\}$ records crossing positions. When $S = \emptyset$, the family is non-intersecting and corresponds to a CPP; when $S \neq \emptyset$, the paths cross and the family carries sign $(-1)^{|S|}$.

The key identification. The crossing set S in a ground-set element $(S, \mathfrak{d}) \in \mathcal{G}_{n,c}$ records exactly *where the lattice paths cross*. The Euler factor $(zq;q)_n$ in the E–H factorisation is the generating function for all possible crossing configurations: each factor $(1-zq^k)$ contributes a choice of whether or not paths cross at level k. The signed sum

$$Q_{n,c}(q) = \sum_{(S,\mathfrak{d})\in\mathcal{G}_{n,c}} (-1)^{|S|}\, q^{\sigma(S)+\mathrm{wt}(\mathfrak{d})}$$

then becomes a statement about cancellation of crossings: positivity means all intersecting families cancel via a sign-reversing involution, leaving only non-intersecting (CPP) contributions.

This chapter also gives the CPP $\leftrightarrow$ ALCD bijection via growth diagrams, which is the combinatorial incarnation of the E–H factorisation and provides the precise description of the ground set $\mathcal{G}_{n,c}$.

Part IV

The Involution

Chapter 8

The Canonical Peel

With the ground set $\mathcal{G}_{n,c}$ and its crossing interpretation in hand, we construct the sign-reversing involution. The local rule at each face of the growth diagram computes the sets A, B, and $A \cap B$: faces with $A \cap B \neq \emptyset$ along the outermost diagonal are the *peel candidates*, where the cascade is actively preventing a crossing. To *peel* means to undo the cascade: let the paths cross, adding one element to S. To *glue* is the reverse: resolve an existing crossing, removing one element from S.

The involution rule is a Franklin-style comparison:

1. Compute the peel level ℓ: the first face with $A \cap B \neq \emptyset$ and strip size = level number.
2. Let $s = \min(S)$ (the smallest existing crossing).
3. If $\ell < s$: peel. If $s < \ell$: glue. If $\ell = s$: fixed point.

Peel is weight-preserving (transfers exactly ℓ between $\sigma(S)$ and $\mathrm{wt}(\mathfrak{d})$) and sign-reversing ($|S|$ changes by ± 1). The comparison with classical LGV (where "swap tails at first intersection" is automatic on a DAG) reveals what the cylinder adds: a cascade direction choice and a weight condition.

Chapter 9

The Recursive Franklin Involution

The single-level Franklin involution of Chapter 8 can leave intersecting fixed points (elements with $S \neq \emptyset$ and $\ell = \min(S)$). The recursive version eliminates them all.

The involution decomposes into n independent levels. At level k, we toggle element k in or out of S:

$$\iota_k(S, w) = \begin{cases} (S \cup \{k\},\ w - k) & \text{if } k \notin S, \\ (S \setminus \{k\},\ w + k) & \text{if } k \in S. \end{cases}$$

This is sign-reversing (since $|S|$ changes by ± 1) and weight-preserving (since $\sigma(S) + w$ is invariant). An element is a level-k fixed point if its partner under ι_k does not exist among the current survivors.

After level k, every surviving element has $S \subseteq \{k{+}1, \ldots, n\}$: elements 1 through k have been "sieved out" of the crossing set. After all n levels, only $S = \emptyset$ remains. Every survivor is non-intersecting and contributes with sign $+1$, giving $Q_{n,c}(q) \in \mathbb{N}[q]$.

Theorem 9.1 (Main result, conditional on $h_m \geq 0$)**.** *Let $c = (c_0, c_1, c_2)$ be a rank-3 profile with $d = c_0 + c_1 + c_2$. If $h_{m,c}(q) \in \mathbb{N}[q]$ for all $0 \leq m \leq n$, then the recursive Franklin involution on $\mathcal{G}_{n,c}$ pairs every element with $S \neq \emptyset$, and $Q_{n,c}(q) \in \mathbb{N}[q]$.*

Computationally verified for all rank-3 profiles with $d \leq 8$ and $n \leq 3$: the condition $h_m \geq 0$ holds whenever $d \not\equiv 0 \pmod 3$.

Appendix A

Computational Evidence

The conjecture has been verified computationally for all rank-3 profiles with $d \leq 7$ and n up to the truncation bound. The Python pipeline (`compute_gk.py`, `verify_conjecture.py`) enumerates cylindric plane partitions as triples $(\nu^{(1)}, \nu^{(2)}, \nu^{(3)})$ satisfying the interlacing conditions, validates against the Borodin product formula, and computes $Q_{n,c}(q)$ via the E–H factorisation.

Key computational findings:

- $d = 2$ gives $Q_n = q^{n^2}$ or $q^{n(n+1)}$ (Rogers–Ramanujan / Andrews–Gordon quadratic forms).
- $h_m = (q)_m \cdot g_m \in \mathbb{N}[q]$ for all tested profiles with $d \not\equiv 0 \pmod 3$.
- $Q_{n,c}(q)$ does *not* decompose as a non-negative sum of Kostka–Foulkes polynomials.
- The q-binomial decomposition $Q_n = \sum_j (-1)^j q^{j(j+1)/2} \begin{bmatrix} n \\ j \end{bmatrix} h_{n-j}$ is verified for all tested cases.

Appendix B

The Cyclic Descent Injection

For $d \not\equiv 0 \pmod 3$, an alternative approach to positivity uses a direct injection from weight-$(w-1)$ to weight-w max-1 cylindric plane partitions. The injection is governed by the *cyclic descent rule*: given the slack vector $s = (s_1, s_2, s_3)$ of a CPP triple, increment the layer l_k where k has $s_k > s_{k-1 \bmod 3}$ with the smallest predecessor. This rule is verified for all profiles with $d = 1, 2, 4, 5, 7, 8$ (126 profiles total). For $d \equiv 0 \pmod 3$, the balanced slack $(d/3, d/3, d/3)$ has no cyclic descent, and the injection fails.

Appendix C

The $d \equiv 0 \pmod 3$ Obstruction

For $d = 3$ (profile $c = (2, 1, 0)$), the polynomial h_1 has negative coefficients:

$$h_1 = 2q + q^2 + q^5 - q^6 + q^8 - q^9 + \cdots .$$

When $h_m \notin \mathbb{N}[q]$, the Franklin involution cannot operate at the h_m level (the ground-set elements would have negative multiplicities). The involution for $d \equiv 0 \pmod 3$ must therefore operate at a *deeper* combinatorial level: on individual CPPs and the $(q)_m$ factor simultaneously, rather than on the compressed h_m polynomials.

This parallels the $d \equiv 0 \pmod 3$ obstruction for the cyclic descent injection (Appendix B), where the balanced slack vector has no cyclic descent. Resolving this obstruction is the main open problem.

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